

Step 7 – 9: Secondary Data & Spreadsheet Organization

ساتویں قدم میں ایک قابلِ اعتماد آن لائن مضمون کا ثانوی ڈیٹا ("زیادہ اسکرین ٹائم توجہ کم کرتا ہے") اپنے سروے کے...

CS XI

Chapter 6: Digital Literacy

Lecture 13 of 14

★ Golden

Step 7 – 9: Secondary Data & Spreadsheet Organization ★ Golden

Step 7: Use secondary data

LEARN IT

Students include a reliable online article stating that "**excessive screen time lowers concentration.**" They will **compare** this secondary data with their own primary survey results. If both agree, the conclusion is stronger. If they disagree, the students must think about why (for example, a different age group or country).

Step 8: Organise data in a spreadsheet

LEARN IT

Responses are entered into **MS Excel** or **Google Sheets** to **sort, calculate totals** and **find averages**. Each row is one student; each column is one question.

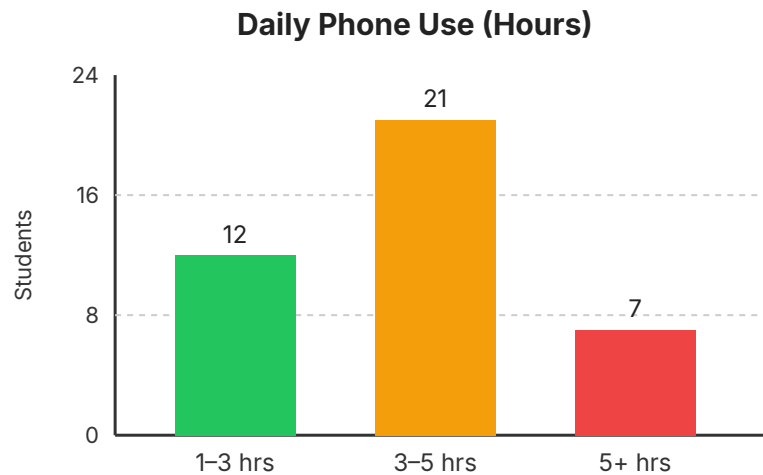
Student	Hours	Study Use	Main Activity	Distracted?
1	3	Yes	Lectures	Sometimes
2	5	No	Social Media	Yes
3	2	Yes	Notes	No

(The full sheet has 40 rows, one per student. The formulas for this table are worked out in topic 6.7.1.)

Step 9: Create charts and graphs

LEARN IT

The spreadsheet data is converted into **visuals** to make it easy to digest. In the source, the "Daily Phone Use (Hours)" bar chart groups the 40 students into three bars: 1–3 hrs, 3–5 hrs and 5+ hrs, with the 3–5 hrs bar the tallest.



Bar chart of daily phone use for 40 students. The counts (12, 21, 7) are estimated from the bar heights in the source chart.

WORKED EXAMPLE

Making the chart in Google Sheets.

Step 1 – Count each group. If the hours are in B2:B41, use `=COUNTIF(B2:B41,"<3")` for under 3 hrs, `=COUNTIFS(B2:B41,"≥3",B2:B41,"<5")` for 3 to under 5 hrs, and `=COUNTIF(B2:B41,"≥5")` for 5 hrs or more. (This gives non-overlapping groups.)

Step 2 – Check the total. $12 + 21 + 7 = 40$ ✓ – every student is counted exactly once.

Step 3 – Percentages. $12 \div 40 = 30\%$; $21 \div 40 = 52.5\%$; $7 \div 40 = 17.5\%$. Total = 100% ✓.

Step 4 – Insert chart. Select the group labels and counts → Insert → Chart → choose *Column chart*. Add the title "Daily Phone Use (Hours)" and label the axes.

Step 5 – Choose the right chart. Bar/column chart to compare groups; pie chart to show parts of a whole (e.g. % in each group); line chart to show change over time.

REMEMBER IT

Chart chooser: **Bar = Compare**, **Pie = Parts of a whole**, **Line = Change over time**.

COMMON MISTAKE

Mistake: Using overlapping groups like "1–3" and "3–5": a student who uses the phone for exactly 3 hours could be counted twice or put in the wrong bar. **Fix:** Define groups clearly, e.g. "under 3", "3 to under 5", "5 or more", and check that the group counts add up to the sample size.

ساتویں قدم میں ایک قابلِ اعتماد آن لائن مضمون کا ثانوی ڈیٹا ("زیادہ اسکرین ٹائم توجہ کم کرتا ہے") اپنے سروے کے بنیادی ڈیٹا سے موازنہ کرنے کے لیے لیا جاتا ہے۔ آٹھویں قدم میں جوابات ایکسل یا گوگل شیٹس میں درج کر کے ترتیب، مجموعہ اور اوسط نکالی جاتی ہے۔ نویں قدم میں اسپریڈشیٹ کے ڈیٹا سے چارٹ اور گراف بنائے جاتے ہیں تاکہ نتائج آسانی سے سمجھ آئیں۔

EXAM TIP

You may get the spreadsheet table and be asked to "write formulas to find the total and average hours" or "count students who use the phone for study". Write the full formula with =, the correct range, and the result.

CHECK YOURSELF

1. What is the secondary data used in Step 7?
2. Which formula counts how many students answered "Yes" in column C, rows 2 to 41?
3. Which type of chart is best for comparing the three hour-groups?

Answers: 1. A reliable online article stating "excessive screen time lowers concentration". 2. =COUNTIF(C2:C41, "Yes") . 3. A bar (column) chart.

Extra board practice

These MCQs and questions are extra to the textbook. Close the notes, attempt them, then open the key.

MCQ 1. The digital inquiry project applies:

- a. All of chapter 6 in order
- b. Only careers
- c. Only NAND
- d. Only OSI

MCQ 2. Advanced search helps you:

- a. Narrow secondary sources (quotes, site, date)
- b. Hack Wi-Fi
- c. Draw K-maps
- d. Sort in $O(1)$

MCQ 3. Primary collection in the project is often:

- a. A class survey
- b. A CPU design
- c. A TCP stack
- d. A waterfall contract

MCQ 4. You organise results in a:

- a. Spreadsheet + charts
- b. Random WhatsApp dump only
- c. NAND netlist
- d. Figma only

MCQ 5. The artefact is:

- a. The final digital product that answers the question
- b. The first brainstorm sticky
- c. The career list
- d. The teacher's attendance

MCQ 6. A conclusion should:

- a. Answer the question and admit limits
- b. Ignore the data
- c. Name and shame students
- d. Invent $n=1$ million

Extra short questions

1. List the project steps in order (grouped as in the book).
2. What is a digital artefact here?
3. Why include secondary if you surveyed?
4. One sentence of a good limitation.

Extra long question

1. Outline a 12-step inquiry: "Does extra screen time lower concentration in our class?" Name tools and the artefact. ★ Golden.

Extra answer key

MCQs: 1. A 2. A 3. A 4. A 5. A 6. A

Short

1. Search/method; survey/primary; secondary+sheet; analyse/conclude; artefact.
2. Slides, infographic, short video, site — that answers the inquiry question.
3. To compare your small sample with a published claim.
4. $n=40$ in one college, so we cannot speak for all Sindh.

Long

1. Steps 1–3 search; 4–6 survey; 7–9 PBS/article + sheet; 10–12 charts, conclusion, artefact (infographic+spreadsheet). Ethics. One expected chart: hours vs self-reported focus.

Next lecture: Step 10 – 12: Data Analysis, Conclusion & Artefact